11-Neuronal-Networks

Note: $\hat{y} = Y_*$ (target of dataset)

1. Para

$$X = \begin{pmatrix} -2 \\ 0 \\ 1.5 \end{pmatrix},$$

calcule $\mathrm{ReLU}(X)$, $\sigma(X)$ para $\sigma(x) = \frac{1}{1+e^{-x}}$, $\tanh(X)$ y las derivadas correspondientes de cada función de activación evaluadas en las entradas dadas. Para ReLU, tome $\mathrm{ReLU}'(0) = 0$.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\frac{d}{dx}\tanh(x) = \frac{[e^x + e^{-x}]^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= 1 - \left[\frac{e^x - e^{-x}}{e^x + e^{-x}}\right]^2$$

$$= 1 - \tanh^2(x)$$

CHECK NOTEBOOK: 11-nn-computations.ipynb

2. Considere la red neuronal con una capa oculta explícita H:

$$Z_1 = W_1 X + B_1, \qquad H = \mathrm{ReLU}(Z_1), \qquad Y = W_2 H + B_2,$$

donde

$$X = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \qquad W_1 = \begin{pmatrix} 1 & 0 \\ 2 & -1 \\ -1 & 1 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix},$$

$$W_2 = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 3 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

Aquí H es la capa oculta. Calcule Z_1, la capa oculta H y la salida final Y.

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3. Sea

$$Y = \begin{pmatrix} 1.2 \\ -0.4 \\ 2.0 \end{pmatrix}, \qquad Y_* = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \qquad E = \frac{1}{2}\|Y - Y_*\|^2.$$

Calcule el error E y el vector $\nabla_Y E$.

$$E = \frac{1}{2}\|Y - \hat{Y}\|^2 = \frac{1}{2}\sum_{i=1}^{3}(Y_i - \hat{Y}_i)^2$$

$$\frac{dE}{\partial Y_i} = \frac{1}{2}\left[2(Y_i - \hat{Y}_i)\right] = Y_i - \hat{Y}_i$$

$$\Rightarrow \nabla_Y E = \left[\frac{\partial E}{\partial Y_1}, \frac{\partial E}{\partial Y_2}, \frac{\partial E}{\partial Y_3}\right]$$

$$= \left[Y_1 - \hat{Y}_1, Y_2 - \hat{Y}_2, Y_3 - \hat{Y}_3\right]$$

$$\nabla_Y E = Y - \hat{Y}$$

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4. Para una capa con activación identidad, use

$$W = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \end{pmatrix}, \qquad X = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \qquad \nabla_Y E = \begin{pmatrix} 4 \\ -1 \end{pmatrix}.$$

Calcule $\nabla_W E$, $\nabla_B E$ y $\nabla_X E$.

$$\hat{Y} = WX + B$$

$$E = \frac{1}{2} \| \hat{y} - y \|^2$$

$$\mathbb{R}^{2\times 3} \ni \frac{dE}{\partial w} = [dE/dw_{ij}]$$

(CHAIN) $w_{ij} \rightarrow \hat{y} \rightarrow E$

- $\frac{dE}{\partial w_{ij}} = \frac{dE}{d\hat{y}} \frac{d\hat{y}}{\partial w_{ij}}$

  $\mathbb{R}^{1\times 2}$ $\mathbb{R}^{2\times 1}$

$$\frac{dE}{\partial \hat{y}} = \hat{y} - y = \frac{\partial E}{\partial y}$$

$$\frac{d\hat{y}_k}{dw_{ij}} = \frac{d}{dw_{ij}} \sum_{I=1}^{3} w_{kI} x_I + b_k = \begin{cases} 0, & k \neq i \\ x_j, & k = i \end{cases}$$

- $\frac{d\hat{y}}{\partial w_{11}} = [\overset{k=1}{x_1} \quad \overset{k=2}{0}]$ $\quad \frac{d\hat{y}}{\partial z_1} = [0, x_1]$
- $\frac{d\hat{y}}{\partial w_{12}} = [\overset{k=1}{x_2} \quad \overset{k=2}{0}]$ $\quad \frac{d\hat{y}}{\partial z_2} = [0, x_2]$
- $\frac{d\hat{y}}{\partial w_{13}} = [x_3 \quad 0]$ $\quad \frac{d\hat{y}}{\partial z_3} = [0, x_3]$

$$\frac{\partial E}{\partial w_{1j}} = \left(\frac{dE}{dy}\right)_1 x_j \qquad \frac{\partial E}{\partial w_{2j}} = \left(\frac{dE}{dy}\right)_2 x_j$$

column

$$\frac{dE}{\partial w} = \begin{bmatrix} \frac{dE}{dy} x_1 & \frac{\partial E}{\partial y} x_2 & \frac{\partial E}{\partial y} x_3 \end{bmatrix}$$

$$\frac{\partial E}{\partial w} = \begin{bmatrix} \left(\frac{dE}{dy}\right)_1 x_1 & \left(\frac{\partial E}{\partial y}\right)_1 x_2 & \left(\frac{\partial E}{\partial y}\right)_1 x_3 \\ \left(\frac{\partial E}{\partial y}\right)_2 x_1 & \left(\frac{\partial E}{\partial y}\right)_2 x_2 & \left(\frac{\partial E}{\partial y}\right)_2 x_2 \end{bmatrix}$$

$$= \frac{dE}{\partial y} \quad (x_1, x_2, x_3) \in \mathbb{R}^{2\times 3}$$

$$2\times 1 \qquad x^T \quad 1\times 3$$

- $\nabla_\beta E$ (CHAIN) $\beta \to y \to E$

know

$$\frac{\partial E}{\partial \beta} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial \beta} \in \mathbb{R}^{1\times 2}$$

$$\mathbb{R}^{1\times 2} \quad \mathbb{R}^{2\times 2}$$

$$\frac{d\,y_i}{\partial \beta_j} = \frac{d}{\partial \beta_j} \sum_{k=1}^{n} w_{ik} x_k + b_k = \delta_{kj}$$

$$\frac{\partial y}{\partial \beta} = I_{2\times 2}$$

$$\Rightarrow \frac{d\,E}{\partial \beta} = \frac{d\,E}{\partial y} \cdot I = \frac{d\,E}{\partial y}$$

- $\nabla_x E$ (CHAIN) $X \rightarrow Y \rightarrow E$

$$\frac{\partial E}{\partial x} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial x}$$

$$\in \mathbb{R}^{1\times 2} \qquad \mathbb{R}^{2\times 3}$$

$$\frac{\partial y}{\partial x}_{ij} = \frac{\partial y_i}{\partial x_j}$$

$$\frac{\partial y_i}{\partial x_j} = \frac{\partial}{\partial x_j} \sum_{k=1}^{3} w_{ik} x_k + b_i$$

$$= w_{ij}$$

$$\frac{\partial y}{\partial x} = w$$

$$\frac{\partial E}{\partial x} = \frac{\partial E}{\partial y} W$$

CHECK NOTEBOOK: 11-nn-computations.ipynb

5. En una capa de activación ReLU, se tiene

$$X = \begin{pmatrix} -3 \\ 0.5 \\ 2 \\ -0.2 \end{pmatrix}, \qquad \nabla_Y E = \begin{pmatrix} 1 \\ -4 \\ 0.5 \\ 3 \end{pmatrix}.$$

Calcule $Y = \text{ReLU}(X)$ y luego calcule $\nabla_X E$ usando el producto de Hadamard.

$Y = \text{ReLU}(x)$

$E = \frac{1}{2} \| y_* - y \|^2$

- $\nabla_x E = \left(\frac{d E}{\partial x_i} \right)_{i=1-4}$

(CHAIN) $x_i \rightarrow y \rightarrow E$ $\qquad \mathbb{R}^{1\times 4}$ $\qquad \mathbb{R}^{4\times 4}$

$$\frac{\partial E}{\partial x_i} = \underbrace{\frac{\partial E}{\partial y}}_{\text{Known}} \cdot \frac{\partial y}{\partial x_i} \Rightarrow \frac{d E}{\partial x} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial x}$$

$\mathbb{R}^{1\times 4}$ $\qquad \mathbb{R}^{4\times 1}$

$$\frac{d y_j}{\partial x_i} = \frac{\partial}{\partial x_i} \text{relu}(x_j) = \begin{cases} 0 & , \ i \neq j \\ 1 & , \ i = j \ , \ x_i > 0 \\ 0 & , \ i = j \ , \ x_i < 0 \end{cases}$$

$$X = \begin{pmatrix} -3 \\ 0.5 \\ 2 \\ -0.2 \end{pmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix}$$

$$\frac{\partial y}{\partial x} = \begin{matrix} i=1 \\ i=2 \\ i=3 \\ i=4 \end{matrix} \begin{bmatrix} j=1 & j=2 & j=3 & j=4 \\ x_1<0 & & & \\ 0 & x_2>0 & & \\ & 1 & x_3>0 & \\ & & 1 & \\ & & & x_4<0 \\ & & & 0 \end{bmatrix}$$

$$\frac{\partial E}{\partial X} = \begin{bmatrix} 0 & \left(\frac{\partial E}{\partial y}\right)_2 & \left(\frac{\partial E}{\partial y}\right)_3 & 0 \end{bmatrix}$$

$$= \frac{\partial E}{\partial y} \odot \operatorname{diag}\left(\frac{\partial y}{\partial x}\right)$$

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6. En una capa de activación logística, considere

$$X = \begin{pmatrix} 0 \\ \ln 3 \\ -\ln 2 \end{pmatrix}, \qquad \nabla_Y E = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}.$$

Calcule $\sigma(X)$, $\sigma'(X)$ y $\nabla_X E$, donde $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

- $\sigma(x) = \frac{1}{1+e^{-x}}$

$$\theta(X) = \begin{bmatrix} \sigma(x_1) \\ \sigma(x_2) \\ \sigma(x_3) \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/(1+1/3) \\ 1/(1+2) \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 \\ 3/4 \\ 1/3 \end{bmatrix}$$

- $\theta'(x) = \sigma(x)(1 - \theta(x))$

$$= \begin{bmatrix} 1/2 \\ 3/4 \\ 1/3 \end{bmatrix} \odot \begin{bmatrix} 1/2 \\ 1/4 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 3/16 \\ 2/9 \end{bmatrix}$$

- $\nabla_x E$

$$Y = \theta(X)$$

$$E = \frac{1}{2}\|Y - Y_*\|^2$$

(CHAIN) $X \rightarrow Y \rightarrow E$

$$\nabla_x E = \left(\frac{\partial E}{\partial x_i}\right)_{i=1-3}$$

KNOWN

$$\nabla_x E = \frac{\partial E}{\partial x} = \frac{\partial E}{\partial y} \frac{\partial Y}{\partial x}$$

$\mathbb{R}^{1\times 3}$ $\mathbb{R}^{3\times 3}$

$$\frac{\partial y_j}{\partial x_i} = \frac{\partial}{\partial x_i}\,\sigma(x_j) = \sigma'(x_j)\,\delta_{ij}$$

$$= \sigma(x_j)(1-\sigma(x_j))\,\delta_{ij}$$

$j=1$ $j=2$ $j=3$

$$\frac{\partial y}{\partial x} = \begin{matrix} i=1 \\ i=2 \\ i=3 \end{matrix} \begin{pmatrix} \sigma(x_1)(1-\sigma(x_1)) & 0 & 0 \\ 0 & \sigma(x_2)(1-\sigma(x_2)) & 0 \\ 0 & 0 & \sigma(x_3)(1-\sigma(x_3)) \end{pmatrix}$$

$$= \text{Diag}\left(\sigma(x_i)(1-\sigma(x_i))\right)$$

$$\nabla_x E = \frac{\partial E}{\partial y}\,\text{Diag}\left(\sigma(x_i)(1-\sigma(x_i))\right)$$

$$= \frac{\partial E}{\partial y} \odot \text{Diag}\left(\sigma(x_i)(1-\sigma(x_i))\right)$$

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7. Realice un paso de descenso del gradiente para una red con una capa escondida:

$$Z_1 = W_1 X + B_1, \qquad H = \text{ReLU}(Z_1), \qquad Y = W_2 H + B_2.$$

Use

$$X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad W_1 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

$$W_2 = \begin{pmatrix} 2 & -1 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 0 \end{pmatrix}, \qquad Y_* = \begin{pmatrix} 1 \end{pmatrix}, \qquad \eta = 0.1.$$

Para $E = \frac{1}{2}\|Y - Y_*\|^2$, calcule Z_1, H, Y, los gradientes $\nabla_{W_2} E$, $\nabla_{B_2} E$, $\nabla_{W_1} E$ y $\nabla_{B_1} E$. Luego calcule $W_{1,\text{nuevo}}$, $B_{1,\text{nuevo}}$, $W_{2,\text{nuevo}}$ y $B_{2,\text{nuevo}}$.

$$Z_1 = W_1 X + B_1 \qquad (2\times2)\ (2\times1)\ (2\times1)$$

FORWARD ↓

$$H = \text{RELU}(Z_1)$$

$$Y = W_2 H + B_2 \in \mathbb{R}^{1\times1} \qquad (1\times2)\ (2\times1)\ (1\times1)$$

[PARAMETERS OF NN]

$$\theta = \text{FLAT}(W_1, B_1, W_2, B_2)$$

$$\theta_{k+1} = \theta_k + \alpha_k p_k \,, \quad p_k = -\nabla_\theta E(\theta_k)$$

$$\alpha_k = \eta = 0.1$$

- $\nabla_\theta E = (\nabla_{W_1} E, \nabla_{B_1} E, \nabla_{W_2} E, \nabla_{B_2} E)$

- $\theta = B_2 \qquad \nabla_{B_2} E = \left(\frac{\partial E}{\partial B_2}\right)$

(CHAIN) $\qquad B_2 \to Y$

$$\frac{\partial E}{\partial B_2} = \frac{\partial E}{\partial Y} \cdot \frac{\partial Y}{\partial B_2}$$

- $\frac{\partial E}{\partial Y} = Y - Y_* \in \mathbb{R}^{1\times1} \qquad (1)$

- $\frac{\partial Y}{\partial B_2} = 1 \in \mathbb{R}$

$$\frac{\partial E}{\partial B_2} = Y - Y_* \qquad (2)$$

- $\theta = W_2$ $\qquad \nabla_{W_2} E = \left(\frac{\partial E}{\partial w_{2ij}} \right)_{\substack{i=1 \\ j=1-2}} \in \mathbb{R}^{2\times 1}$

(CHAIN) $\qquad W_2 \to Y \to E$

$$\frac{\partial E}{\partial w_{2ij}} = \underbrace{\frac{\partial E}{\partial Y}}_{\text{KNOWN by (1)}} \cdot \underbrace{\frac{\partial Y}{\partial w_{2ij}}}_{1\times 1}$$

$$\frac{\partial Y}{\partial w_{2ij}} = \frac{\partial}{\partial w_{2ij}} \left(\sum_{m=1}^{2} w_{2im} H_m + B_2 \right)$$

$$= H_j \in \mathbb{R}$$

$$\frac{\partial E}{\partial w_{2j}} = \frac{\partial E}{\partial Y} H_j \in \mathbb{R}$$

$$\nabla_{w_2} E = \frac{\partial E}{\partial w_2} = \underbrace{\frac{\partial E}{\partial Y}}_{1\times 1} \underbrace{H}_{2\times 1} \in \mathbb{R}^{2\times 1} \qquad (3)$$

- $\theta = B_1$ $\qquad \nabla_{B_1} E = \left(\frac{\partial E}{\partial B_{1j}} \right)_{j=1-2} \in \mathbb{R}^{1\times 2}$

$$B_1 \to Z_1 \to H \to Y \to E$$

$$\frac{\partial E}{\partial B_1} = \underbrace{\frac{\partial E}{\partial Y}}_{\text{KNOWN by (1)}} \cdot \frac{\partial Y}{\partial H} \cdot \frac{\partial H}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial B_1} \in \mathbb{R}$$

$\frac{\partial E}{\partial B_{1j}}$ $\quad$ $\frac{\partial E}{\partial y}$ $\quad$ $\frac{\partial y}{\partial H}$ $\quad$ $\frac{\partial H}{\partial z_1}$ $\quad$ $\frac{\partial z_1}{\partial B_{1j}}$

1×1 $\quad$ 1×2 $\quad$ 2×2 $\quad$ 2×1

- $\frac{\partial y}{\partial H} = w_2 \in \mathbb{R}^{1\times2}$ (4)

- $\frac{\partial H_i}{\partial z_{1j}} = \delta_{ij} \cdot 1\{z_{1j} > 0\}$ (5)

- $\frac{\partial z_{1i}}{\partial B_{1j}} = \delta_{ij}$ $\quad \frac{\partial z_1}{\partial B_{1j}} = \begin{bmatrix} \delta_{1j} \\ \delta_{2j} \end{bmatrix}$

$$\nabla_{B_1} E = \begin{bmatrix} \frac{\partial E}{\partial B_{11}} \\ \frac{\partial E}{\partial B_{12}} \end{bmatrix}^T = \frac{\partial E}{\partial y} \frac{\partial y}{\partial H} \frac{\partial H}{\partial z_1} \begin{bmatrix} \frac{\partial z_1}{\partial B_{11}} & \frac{\partial z_1}{\partial B_{12}} \\ j=1 & j=2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

I

$$\nabla_{B_1} E = (y - y_t) \underset{1\times1}{} \underset{1\times2}{w_2} \underset{2\times2}{\begin{bmatrix} 1\{z_{11}>0\} & 0 \\ 0 & 1\{z_{12}>0\} \end{bmatrix}} \in \mathbb{R}^{1\times2}$$

- $\theta = W_1$

$$\nabla_{W_1} E = \left(\frac{\partial E}{W_{1\,ij}} \right)_{\substack{i=1-2 \\ j=1-2}} \in \mathbb{R}^{4\times 1}$$

(CHAIN) $W_1 \rightarrow Z_1 \rightarrow H \rightarrow Y \rightarrow E$

$$\frac{\partial E}{\partial W_{1\,ij}} = \underbrace{\frac{\partial E}{\partial Y}}_{\text{KNOWN BY (1)}} \cdot \underbrace{\frac{\partial Y}{\partial H}}_{\text{KNOWN by (4)}} \cdot \underbrace{\frac{\partial H}{\partial Z_1}}_{\text{KNOWN by (5)}} \frac{\partial Z_1}{\partial W_{1\,ij}}$$

$1\times 1 \quad 1\times 2 \quad 2\times 2 \quad 2\times 1$

$$\frac{\partial Z_{1k}}{\partial W_{1\,ij}} = \frac{\partial}{\partial W_{1\,ij}} \sum_{m=1}^{2} W_{1\,k\,m}\, X_m + B_{1m}$$

$$\frac{\partial Z_{1k}}{\partial W_{1\,ij}} = \delta_{ki}\, X_j$$

$$\frac{\partial Z_1}{\partial W_{1\,ij}} = \begin{bmatrix} \delta_{1i} & X_j \\ \delta_{2i} & X_j \end{bmatrix} \in \mathbb{R}^{2\times 1}$$

$$\frac{\partial E}{\partial W_{1\,ij}} = (Y - Y_t)\, W_2 \begin{bmatrix} 1\,|\,z_{11}>0 & 0 \\ 0 & 1\,|\,z_{12}>0 \end{bmatrix} \begin{bmatrix} \delta_{1i}\, X_j \\ \delta_{2i}\, X_j \end{bmatrix}$$

$1\times 2 \qquad 2\times 2 \qquad 2\times 1$

$$\begin{bmatrix} \mathbb{1}\{z_2 > 0\}\, x_2 \\ 0 \end{bmatrix} w_2 (y - y_2) \Bigg) = \begin{bmatrix} \frac{\partial E}{\partial w_{22}} \\ \frac{\partial E}{\partial w_{21}} \\ \frac{\partial E}{\partial w_{12}} \\ \frac{\partial E}{\partial w_{11}} \end{bmatrix} = \nabla_{w_1} L$$

$$\begin{bmatrix} 1\{z_2 > 0\}\, x_1 \\ 0 \end{bmatrix} w_2 (y - y_2)$$

$$\begin{bmatrix} 0 \\ 1\{z_1 > 0\}\, x_2 \end{bmatrix} w_2 (y - y_1)$$

$$\begin{bmatrix} 0 \\ 1\{z_1 > 0\}\, x_1 \end{bmatrix} w_2 (y - y_1)$$